

40_CS4032 Natural Language Processing — Individual Assignment 02

Module: 40_CS4032 Natural Language Processing

Assignment Title: Offline Intelligent Sinhala Open-Ended Answer Scorer (History of Sri Lanka) using OLLAMA + RAG + Ontology + Agent Architecture

Type: Individual

Total Marks: 100

1. Problem Statement

Develop a **fully offline** intelligent system that **scores Sinhala open-ended answers** for **History of Sri Lanka** questions.

For each question, the system must grade the student answer **out of 20 marks** using a **marking guide**, and produce an **explainable score breakdown** (why marks were awarded/deducted).

The solution must run on **OLLAMA locally** and provide a **Streamlit interface**.

2. Topic Scope (Choose ONE option only)

Select one of the following two history scopes and design your questions, knowledge base, and ontology accordingly.

Option 1: Ancient Sri Lanka (Anuradhapura Period)

Focus areas may include: administration, irrigation/civilization, Buddhism & culture, notable rulers/events.

Option 2: Colonial Sri Lanka (Portuguese → Dutch → British)

Focus areas may include: administration changes, economic systems, social/religious impacts, major events and reforms.

3. Question Set Requirement (Maximum 5 Questions)

Create **up to 5 questions** (maximum).

Each question must:

- Be answered in **Sinhala**
- Be graded **out of 20**
- Have a **clear marking guide** (criteria + mark allocation adding up to 20)

4. System Requirements (Must Have)

Your system must include all of the following:

1. **Offline operation** (no Internet connectivity during execution).
2. **OLLAMA-based scoring** (local model inference).

3. **RAG**: retrieve relevant local content to support scoring.
4. **Ontology component**: represent key concepts/relationships for the chosen history scope and use it in the scoring/explanation process.
5. **Agent-based architecture**: an agentic workflow where responsibilities are separated (e.g., retrieval, checking marking guide coverage, scoring, explanation).
6. **Explainability**:
 - Final score /20
 - Mark breakdown per marking criteria
 - Evidence-based justification (grounded in retrieved context and/or ontology concepts)
7. **Streamlit UI**:
 - Select a question
 - Enter Sinhala answer
 - Display score + breakdown + explanation clearly

5. Documentation Requirement (Mandatory)

A **clear operational flow chart** is compulsory.

Each major component/block shown in the flow chart must be **explained in the report with relevant code snippets extracted from your implementation.**

6. Submission Instructions (Strict)

- **Only ONE PDF document** allowed
- **Not exceeding 15 pages**
- **No ZIP uploads**
- Execution demonstrations must be included as **URL links inside the PDF** (videos must be accessible)
- The video must show the system running **offline**, including scoring for multiple answers.

7. Marking Scheme (100 Marks)

A. Problem Setup & History Scope (10 marks)

- Clear selection of one scope and alignment of the system to it (10)

B. Question Set + Marking Guides (15 marks)

- Up to 5 well-designed questions (5)
- Marking guides are clear, structured, and total 20 marks each (10)

C. RAG Implementation (15 marks)

- Correct offline retrieval pipeline (8)
- Retrieved evidence is used meaningfully in scoring/explanations (7)

D. Ontology (15 marks)

- Quality and relevance of ontology to the selected scope (8)
- Ontology is actually used in grading/explanation (7)

E. Agent-Based Architecture (15 marks)

- Clear agent responsibilities and workflow (8)
- Agent design improves reliability (e.g., checking coverage, consistency) (7)

F. Explainable Scoring Quality (20 marks)

- Accurate/defensible score breakdown per marking criteria (10)
- Clear justification of awarded/deducted marks (10)

G. Streamlit UI & Usability (5 marks)

- Functional and clear interface, outputs presented well (5)

H. Report Compliance + Evidence (5 marks)

- Flow chart + component explanations with code snippets included (3)
- Execution video URL(s) + offline proof included; PDF $\leq$ 15 pages (2)